The Validity Net

Accessibility note: This handout recreates the spatial layout of the original “validity net” concept map. If you use a screen reader, a fully linearized companion version with identical content is available as **validity-net-linear.pdf**.

Measurement Validity

- Measurement error → bad data → worthless results
- Is the IV properly manipulated?
- Is the DV properly measured?
- Are the values we have for **every** measure/behavior correct?
- Observational, Self-report or Trace?
- Primary or Archival data?

Every Characteristic/ Behavior/Procedure is either…

	Constant	Variable
Measured	1	2
Manipulated	3	4

IV causes → DV

- **Temporal Precedence**
- **Reliable statistical relationship**
- **No alternative hypotheses/confounds**

Statistical Conclusion Validity

- IV & DV can’t be causally related if not statistically related
- Statistical significance tests
- Programmatic Research – novel RH tests, replication & convergence

External Validity

Population -- Participant Sampling

Target population	Complete or Purposive
Sampling Frame	Researcher-selected or Self-selected
Selected Sample	Simple or Stratified
Data Sample	Attrition

Setting

- Laboratory, Structured or Field?

Task/Stimulus

- Familiar/Representative or Unfamiliar/Control?

Societal/Temporal

- Relationships among variables change over time in a society

types & their “roles” in a design

Characteristic/Behavior/ Procedure Plays a “Role” in a Study

Causal Variable (IV)	4 – Ongoing Eq
Effect Variable (DV)	2 – Ongoing Eq
Control Constant	1 – Initial Eq 3 – Ongoing Eq
Control Variable	2 – Initial Eq 4 – Ongoing Eq
Confounding Variable	2 – Initial Eq 4 – Ongoing Eq

Choices we make influence Internal and External Validity !!

Participants
representation vs. control

Setting
representation vs. control

Task-Stimulus
representation vs. control

Internal Validity

Design

	BG	WG
True Experiment	✓	✓
Non-experiment	✗	✗

Legend: ✓ = design supports causal interpretation; ✗ = design does not support causal interpretation.

Initial Equivalence – Participant Assignment

- RA of individual participants by the researcher before manipulation of the IV -- best but not a guarantee
- Without proper RA all subject variables are potential confounds
- Subject constants can't be confounding variables
- Subject variables that are equivalent across IV conditions are control variables
- Subject variables that are nonequivalent across IV conditions are confounding variables – even if RA was used (remember RA doesn't always work)

Ongoing Equivalence – Procedural Standardization

- Only the IV is different across IV conditions
- Procedural constants cannot be confounding variables
- Procedural variables that are equivalent across IV conditions are control variables
- Procedural variables that are nonequivalent across IV conditions are confounding variables
- Ongoing equivalence is harder to maintain in field settings
- Ongoing equivalence is harder to maintain during longer procedures